

REPORT

Title

Prolog Program to Suggest Diet Based on Disease

Aim

To implement a **disease-based diet suggestion system using Prolog** with facts and rules.

Problem Statement

Create a Prolog program that stores general dietary recommendations for different diseases. When a disease is given as a query, the program should display recommended foods and foods to avoid.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp8.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp8.pl`.
3. Enter the disease and diet facts.
4. Define a rule to retrieve the diet information.
5. Save the program.
6. Load the file using `[exp8]..`
7. Enter a disease as a query.
8. Observe the recommended foods and foods to avoid.

Program

```
% Disease and recommended diet

diet(diabetes, recommended, [vegetables, whole_grains, legumes]).
diet(diabetes, avoid, [sugary_drinks, sweets, refined_sugar]).

diet(hypertension, recommended, [fruits, vegetables, whole_grains]).
diet(hypertension, avoid, [salty_foods, processed_foods, excess_salt]).
```

```
diet(anemia, recommended, [spinach, beans, lentils, iron_rich_foods]).
diet(anemia, avoid, [excess_tea, excess_coffee]).

diet(obesity, recommended, [vegetables, fruits, whole_grains]).
diet(obesity, avoid, [fast_food, sugary_drinks, fried_foods]).

suggest_diet(Disease, Recommended, Avoid) :-
    diet(Disease, recommended, Recommended),
    diet(Disease, avoid, Avoid).
```

Sample Output

Query:

```
?- [exp8].
true.

?- suggest_diet(diabetes, Recommended, Avoid).
```

Output:

```
Recommended = [vegetables, whole_grains, legumes],
Avoid = [sugary_drinks, sweets, refined_sugar].
```

Another query:

```
?- suggest_diet(hypertension, Recommended, Avoid).
```

Output:

```
Recommended = [fruits, vegetables, whole_grains],
Avoid = [salty_foods, processed_foods, excess_salt].
```

Conclusion

Thus, a **disease-based diet suggestion system** was successfully implemented in **Prolog**. The program retrieves general recommended foods and foods to avoid based on the given disease. These are general educational examples, not personalized medical advice.